# Quotient calibration: validator review correction

**isogeny-labs/crypto · PR 2641 · 2026-10-30**

The projective validator now rejects incomplete coordinates before target arithmetic. **21 focused tests pass. No new target frame, challenge image or inverse divisor has been computed.** The F343/order-373/ kernel-153 inputs, certified anchor and acceptance criteria are unchanged.

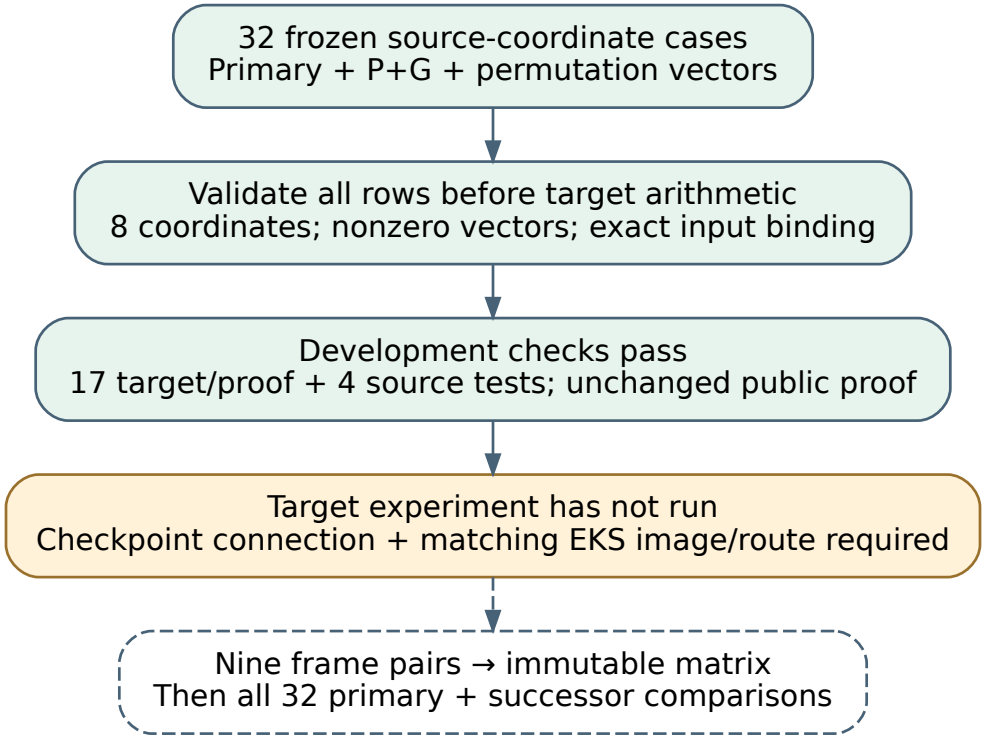


Figure 1: Green: fixed development checks. Amber: unresolved execution prerequisites. Dashed: unexecuted target experiment.

## Correctness and fixed validation

The former zip comparison could accept a one-coordinate prefix. The repair rejects unequal lengths and checks every prediction row before constructing any target divisor. All 32 case IDs, source points and P+G inputs must match. Each primary, successor and control vector has exactly eight field coordinates and is nonzero; field coefficients lie in 0..6.

| Check | Observed result |
|---|---|
| Native focused tests | 8 target components + 9 certificate tests + 4 original source tests pass. Strict all-target Clippy passes. |
| Historical targets | 3 of 3 existing F49 vectors match both existing paths. No new target is evaluated. |
| Public identity proof | IDC1h368ff5f1ba01b556 and the exact integer expansion replay byte-for-byte. No credential/TLS content. |
| Root library checks | Earlier run: 4,845 pass; 17 fail; 121 ignored (absent fixtures). Current locked check stops at an inherited manifest/lock mismatch. No full-suite pass. |

## Execution remains pending

The user approved ongoing same-branch checkpoint uploads on 30 October. The environment still has no configured upload transport; the normal gate must pass before an experiment. The rejected pasted-key process was not retried. The draft plan now follows the repository’s EKS Spot default. A matching target-binary image and custom-command route need qualification; the standard GitHub workflow builds ecbench and is not a substitute.

After access and image qualification: prospective preflight, freeze/commit, nine-point calibration, immutable matrix checkpoint, then all 32 primary and successor comparisons. No refit, point replacement or omitted failure. Inversion, signs, subgroup/Frobenius checks, eight addition/scalar relations, Profile B and a global section identity remain open. No speedup or vulnerability is established.

Sources/evidence: REPORT.md, raw focused logs, validation.json and merge-preservation.json beside this PDF; Rust Iterator::zip primary documentation; Tian arXiv:2012.07173v1 §3.2; Khuri-Makdisi math/0105182 Lemmas 2.1–2.3. Main integration 1733dc59e preserves quotient records and main problem rows. Historical reports and canonical cost graphs are unchanged.